

Uncertainty Modeling of the Transmission Map for Single Image Dehazing

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Abstract—Despite rapid progress of end-to-end optimization for single-image dehazing, a long-standing open problem is the non-homogeneous haze, at the core of the differences between synthetic hazy images and real hazy images. The atmospheric scattering model (ASM) has been widely adopted to model the degradation process of haze images but based on the assumption of homogeneous haze. In realistic scenarios, non-homogeneous haze often makes it more difficult to estimate the transmission map in ASM, resulting in undesired artifacts in the restored images. To address the issue of non-homogeneous haze, we propose to model the uncertainty in the estimation of the transmission map and develop a spatially adaptive learning module for ASM correction. Specifically, we present an approach to enhancing the well-known Dark Channel prior (DCP) by relaxing the constraint with the transmission map in the DCP-net. Assuming the availability of paired training data, we have developed a strategy to address vulnerability in the DCP, leading to a more accurate estimation of the transmission map. Then, we explore the uncertainty between the estimated transmission map and target transmission map (Ground Truth) to reformulate the ASM for the presence of non-homogeneous haze. A robust and accurate estimated transmission map can boost the final dehazing performance of our DCP-net. Experiments on three popular synthetic and real non-homogeneous datasets show that our proposed approach has achieved better results on both synthetic scenes and real non-homogeneous scenes.

Index Terms—Image Dehazing, Uncertainty Modeling, DCP-net, Transmission Map

I. INTRODUCTION

As one of the fundamental problems in low-level vision, single image dehazing aims to restore the haze-free image from the degraded hazy counterparts, attracting a lot of attention among researchers in recent years. Due to the existence of floating particles in the atmosphere, images taken in such scenes could be corrupted by color distortion, blurring, and low contrast. haze scenarios harm the accuracy of many high-level vision tasks, such as object detection [11], [21], [26], [60], segmentation [3], [10], [34], [56], and especially in high-risk scenes, such as autonomous driving [51]. To approximate

the degradation of hazy images, the atmospheric scattering model(ASM) [37] was proposed, which can be formulated as:

$$I(x) = J(x) t(x) + (1 - t(x))A \quad (1)$$

where $I(x)$ denotes the hazy image, $t(x)$ denotes the transmission map, $J(x)$ and A are the radiance of the scene and the global light of the atmosphere, respectively. It is often difficult to recover a clean image from the hazy image directly. Previous work [5], [16], [46], [22], [17], [62], [27] proposed to calculate the clean image using the ASM shown in Eq. (1) with an estimated transmission map by statistical priors [22], [17]. However, these approaches are mainly based on statistical priors [22], [17] which are sometimes unreliable, leading to inaccurate estimation of the transmission map and undesirable dehazing performance.

With the prosperity of convolutional neural networks (CNNs), deep learning-based works [7], [42], [32], [52], [14], [41], [19], [40], [44], [13], [49], [20], [53], [47], [55], [54], [45], [63], [57] have been proposed to dehaze single images, achieving more robust and stable performance than traditional approaches. These learning-based methods can be roughly divided into two branches: the one involved with the atmospheric scattering model [7], [42], [32], [52], [19], [53], [47], [54] and the other without [14], [41], [40], [44], [13], [49], [20]. For the second without ASM, methods such as [14], [41], [40], [55] only regarded the single image dehazing problem as a simple regression problem and trained the networks end-to-end with a large number of synthetic pairs of clean images. However, there exists a large domain gap between synthetic hazy images and real hazy images. Dehazing networks that are solely trained on synthetic images often lead to poor performance in real-world scenes.

For learning-based methods that involve ASM, it is crucial to estimate physical parameters in ASM such as the transmission map $t(x)$, the radiance of the scene $J(x)$, and the global light of the atmosphere A . The most important issue lies in the accurate estimation of the transmission map $t(x)$. Many works have proposed complex and effective CNN-based networks [7], [42], [52], [40], such as FFA-net [40] and training strategies [14], [41], [44], [13], [49], such as the boost strategy [13] and contrastive learning [49] to accurately estimate the transmission map. However, for outdoor scenes, the lack of ground truth in the transmission map is always a big hindrance for ASM-based methods. In [22], Dark Channel Prior (DCP) was proposed by He *et al.* and a rough estimate of the transmission map can be regarded as guidance. In DCP, the transmission map is assumed to be constant in the local patch.

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Such an assumption is too limited and would fail in some edge cases, resulting in an incorrect estimate of the transmission map. Therefore, we propose to solve this problem by refining the DCP on the premise that paired data are available. Using the clean-haze image pairs, a better transmission map can be obtained and can provide better guidance when estimating the transmission map.

On the other hand, the DCP-calculated transmission map or the proposed approach only provides a rough label to estimate the CNN transmission map since there is still a significant gap between the DCP-calculated transmission map and the real ones, which can hardly be obtained due to the nonlinear dependency on the depth of the unknown scene. Therefore, we propose to explicitly model the uncertainty of the transmission map data in deep neural networks. More specifically, we first reformulate the ASM for non-homogeneous haze by extending the scattering coefficient into a spatially varying function. To decouple the nonlinear dependency between the scattering coefficient and scene depth, we take a linear approximation strategy, which facilitates uncertainty modeling on the transmission map. Inspired by our previous work [38], we have designed a new uncertainty-driven loss to adaptively characterize confidence in the transmission map.

The main contribution of this paper can be summarized as follows.

- In [22], He *et al.* proposed an assumption that the transmission map in a local patch is constant, while we observe that this assumption could fail in some edge cases. Based on the observation, we propose to refine the dark channel prior on the premise that paired data are available. Using the clean-haze image pairs, a better estimation of the transmission map can be obtained, as a better guide when estimating the transmission map.
- Although a better transmission map can be calculated by refining the DCP, there is still uncertainty in estimating the transmission map due to the complex atmospheric environment. To our knowledge, we are the first to explore uncertainty in the transmission map estimation process, achieving much better performance than not.
- For non-homogeneous hazy images, we propose to reformulate the ASM with a linear combination of the two changing factors, the scattering coefficient and the depth, to better handle the complex and difficult estimation of the transmission map. The final dehazing performance shows that our method is effective.
- Extensive experimental results on synthetic and real datasets demonstrate that the proposed method performs better than current state-of-the-art single image dehazing methods, including AECR-Net [49], UHD [58], and Dehamer [20].

II. RELATED WORK

A. Single Image Dehazing

Single image dehazing aims to generate haze-free images from observed hazy ones. Approaches to dehazing can be roughly divided into two categories, which are prior-based methods [16], [46], [22], [17], [5], [62] and learning-based

methods [7], [42], [32], [52], [14], [41], [19], [40], [44], [13], [49], [20].

In the early stage of single image dehazing, most of the works [16], [46], [22], [17], [5], [62] are based on the physical atmospheric scattering model (ASM). Many different image statistics priors [22], [17] from empirical observation as additional constraints have been proposed and used, such as contrast maximization [46], dark channel prior [22], and *et al.* trying to retrieve lost information during the hazing degradation process. Dark Channel prior [22] is one of the most famous priors for single image dehazing. The DCP prior is based on the assumption that the scene depth is constant in the local patch and that the pixels in clean local regions have low intensity in at least one color channel. With this assumption, DCP can provide a fast approach to estimate the transmission map from observed hazy images. Following DCP, a variety of priors for image dehazing have been proposed, such as color-line prior [17], non-local path prior [5] and *et al.* These priors and assumptions are effective in the most general scenes or atmospheric conditions and would fail in some real challenging cases. For example, in DCP [22], the assumption that the scene depth is constant in the local patch could be broken in the edge areas.

With the prevailing success of deep learning in computer vision, more and more approaches [7], [42], [32], [52], [14], [41], [19], [40], [44], [13], [49], [20] based on deep convolution networks have been proposed for the single image dehazing. Some of those deep-dehazing models are based on the ASM with learned deep networks for estimation of the crucial variables in Eq. (1), achieving promising performance. In [7], Cai *et al.* are the first to use deep networks to estimate the transmission map with greater precision. Ren *et al.* [42] proposed a multiscale network to improve the performance of dehazing from coarse to fine. Li *et al.* [32] reformulated the ASM shown in Eq. 1, integrating the transmission map and atmospheric light into one variable, obtaining better results. In [52], Zhang *et al.* proposed a densely connected pyramid neural network and a new edge-preserving loss function for the dehazing task. Golts *et al.* [19] directly leveraged convolution neural networks with the dark channel prior and proposed a completely unsupervised training strategy by minimizing the energy function of the dark channel prior. Other deep dehazing approaches [14], [41], [40], [44], [13], [49], [20] that are trained in an end-to-end manner, simply regard the dehazing task as a regression problem, ignoring the physical model of the hazy images. The CycleGAN [61] framework proposed for style transformation has been adopted by Engin *et al.* [14] for the dehazing task by transforming clean images into hazing images and transforming hazing images into clean ones. Qin *et al.* [40] introduced the attention structure into the dehazing network and achieved state-of-the-art performance at that time, and the proposed network is chosen as our backbone network. Some other work developed different training strategies for image dehazing, such as the boosting strategy [13] and contrastive learning [49].

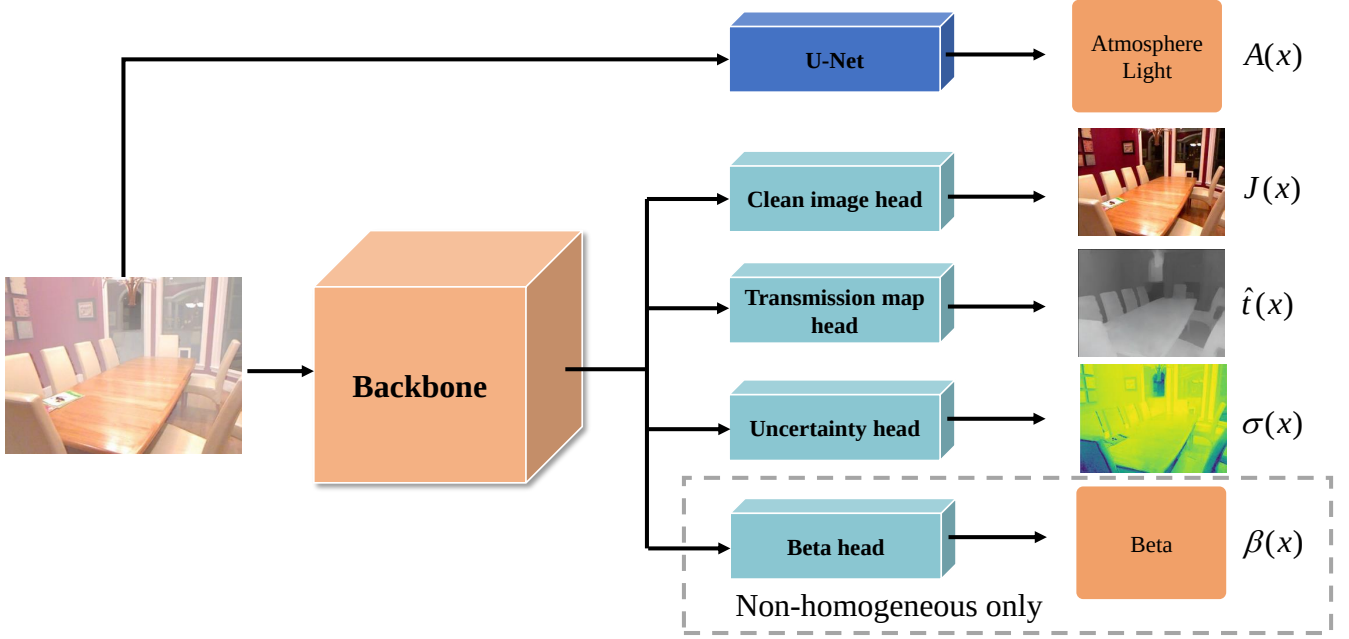


Fig. 1: The overall framework of the proposed approach, which is based on a Physics-compatible model for single image dehazing.

B. Uncertainty in Computer Vision

The uncertainty modeling [18], [48], [6] has been studied for classification and regression tasks for a long time. Recently, uncertainty has been introduced into deep learning with the expectation that improving the performance and robustness of deep neural networks will benefit many different computer vision tasks [28], [4], [8], [38], such as high-level tasks such as image segmentation [28], [4] and face recognition [8], [43], and low-level tasks such as image super-resolution [30], [38], [15], [39]. There are two main categories [12] in uncertainty modeling. The first is the model/epistemic uncertainty, which describes how much the model is uncertain about its predictions. Another type is data/aleatoric uncertainty, which refers to the noise inherent in the observation data. For high-level tasks, Kendall *et al.* [28] proposed a novel framework that introduces both epistemic uncertainty and aleatoric uncertainty for semantic segmentation and depth regression tasks in a deep learning approach. In [8], the uncertainty of face recognition data has been fully studied with improved performance and robustness. Regarding low-level tasks, Ning *et al.* [38] explored the uncertainty of reconstruction difficulty and proposed an adaptive weighted loss for single image super-resolution (SR) based on the estimated uncertainty. In [15], they proposed to estimate the uncertainty in kernel estimation for blind SR, achieving better results for blind SR. For single-image dehazing, one of the challenges is to estimate the transmission map precisely under the atmospheric scattering model. Despite the fact that a rough estimate of the transmission map can be obtained via DCP, there is still a major gap between the calculated and the real ones. To address this issue, we propose to model the uncertainty of the transmission map calculated using our proposed approach. UDN [24] utilized the uncertainty of the feature map as confidence values. In other

words, the uncertainty here is more like a weight. Moreover, the method did not follow the Atmospheric Scattering Model and directly produced the dazed result. The uncertainty is modeled in the feature map field, which is different from ours.

III. DEHAZING BY DCP-NET

In this section, we first briefly illustrate the dark channel prior and show the failed cases in DCP and present how to improve it on the premise that paired data are available. Then, we will introduce the proposed novel framework that explores the uncertainty in the estimation of the transmission map, boosting the dehazing performance.

A. Background

1) *Dark Channel Prior*: The dark channel prior [22] describes an image statistical property that in the haze-free image, at least one color channel has very low intensity at some pixels in a local patch. In other words, the darkest pixel across all color channels in every small patch of a haze-free image is close to zero, which could be expressed by:

$$J^{dark}(\mathbf{x}) = \min_{c \in r, g, b} (\min_{\mathbf{y} \in \Omega(\mathbf{x})} (J^c(\mathbf{y}))), \quad (2)$$

where $\mathbf{J}$ denotes the clean image, J^c is a color channel of $\mathbf{J}$, $\Omega(\mathbf{x})$ denotes the small patch centered around $\mathbf{x}$, and $J^{dark}(\mathbf{x})$ represents the dark channel of $\mathbf{J}$. This statistic property [22] comes mainly from three factors: a) shadows; b) colorful objects, any objects where one color is dominant and other colors are close to zero; c) dark objects or surfaces.

In [22], He *et al.* first assumed that atmospheric light A is given and reformulated Eq. (1) as followed:

$$\frac{I^c(\mathbf{x})}{A^c} = t(\mathbf{x}) \frac{J^c(\mathbf{x})}{A^c} + 1 - t(\mathbf{x}). \quad (3)$$

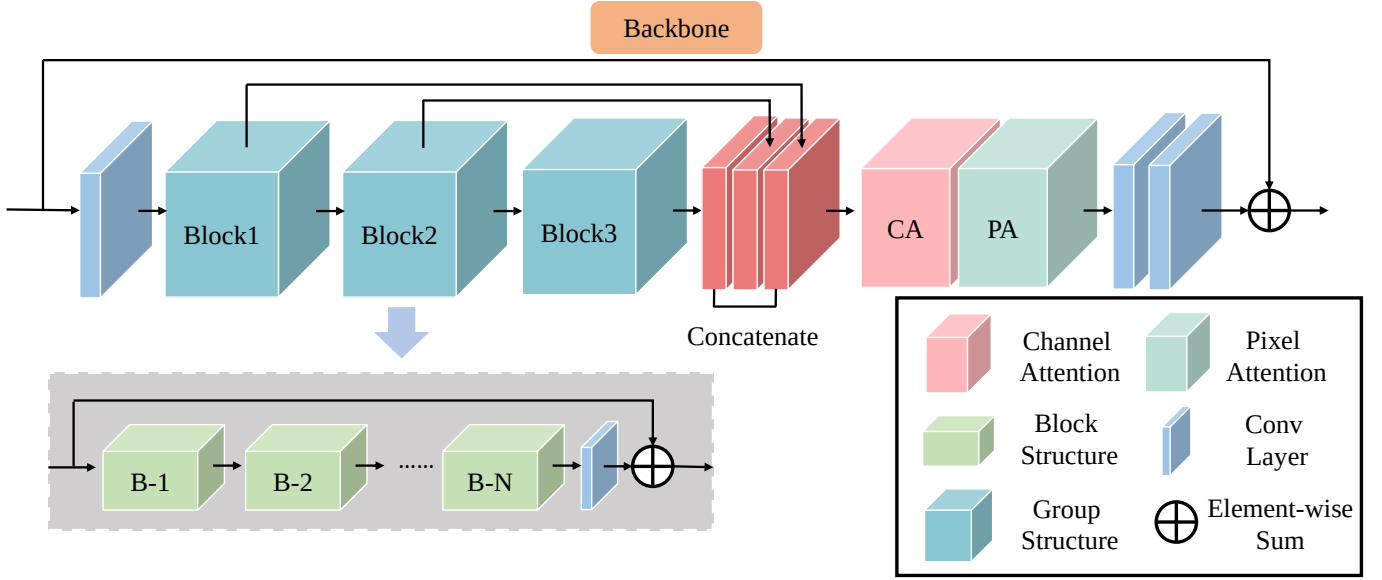


Fig. 2: The backbone network in our framework is FFA-Net [40]. More details of the backbone network can be seen in [40].

They also assumed that the transmission map in a small local patch $\Omega(\mathbf{x})$ is constant. The transmission map of a single patch is denoted as $\tilde{t}(\mathbf{x})$. Then, they implement the Min operation among three color channels in the local patch on both sides of (3):

$$\min_{\mathbf{y} \in \Omega(\mathbf{x})} \left(\min_c \frac{I^c(\mathbf{y})}{A^c} \right) = \tilde{t}(\mathbf{x}) \min_{\mathbf{y} \in \Omega(\mathbf{x})} \left(\min_c \frac{J^c(\mathbf{y})}{A^c} \right) + 1 - \tilde{t}(\mathbf{x}). \quad (4)$$

With the statistic property of dark channel prior, a rough prediction of the transmission map can be obtained by integrating Eq. (2) and Eq. (4):

$$\tilde{t}(\mathbf{x}) = 1 - \omega * \min_{\mathbf{y} \in \Omega(\mathbf{x})} \left(\min_c \frac{I^c(\mathbf{y})}{A^c} \right), \quad (5)$$

where ω is to keep a very small amount of haze for distant objects and is suggested to be 0.95.

$$\begin{bmatrix} 0.8 & 0.8 & 0.8 \\ 0.8 & 0.7 & 0.8 \\ 0.8 & 0.6 & 0.8 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \frac{1}{3} & 1 \\ 1 & \frac{4}{5} & 1 \end{bmatrix} \times \begin{bmatrix} 0.8 & 0.8 & 0.8 \\ 0.8 & 0.1 & 0.8 \\ 0.8 & 0.5 & 0.8 \end{bmatrix} + 1 - \begin{bmatrix} 1 & 1 & 1 \\ 1 & \frac{1}{3} & 1 \\ 1 & \frac{4}{5} & 1 \end{bmatrix}$$

$I(\mathbf{x}) \qquad t(\mathbf{x}) \qquad J(\mathbf{x}) \qquad t(\mathbf{x})$

Fig. 3: A simple toy example of the vulnerability in DCP as illustrated in Section III-A2.

2) Vulnerability in Calculation of Transmission Map:

Although the calculation of the transmission map by Eq. (5) is simple and effective, the assumption that the transmission map in a small local patch is constant could be too strict and could not be satisfied in edge areas. For example, assuming that $\tilde{t}(\mathbf{x})$ is constant in a local patch, the Eq. (4) can be reformulated as :

$$\mathbf{M}_2 = a \mathbf{M}_1 + (1 - a), \quad (6)$$

where a denotes $\tilde{t}(\mathbf{x})$ which is a constant, and $\mathbf{M}_1$ and $\mathbf{M}_2$ denotes the corresponding matrix in Eq. (4). Then, Eq. (6)

is equal to a linear transformation, since a is a constant. Therefore, the index of the darkest pixel in $I^c(\mathbf{y})$ corresponds only to the index of the darkest pixel in $J^c(\mathbf{y})$, which can be described as follows:

$$\begin{aligned} (i_I, j_I, c_I) &= \operatorname{argmin}_{i,j \in \Omega(\mathbf{x})} \left(\operatorname{argmin}_c \frac{I^c(i,j)}{A^c} \right), \\ (i_J, j_J, c_J) &= \operatorname{argmin}_{i,j \in \Omega(\mathbf{x})} \left(\operatorname{argmin}_c \frac{J^c(i,j)}{A^c} \right), \\ (i_I, j_I, c_I) &= (i_J, j_J, c_J). \end{aligned} \quad (7)$$

In other words, the position i_I, j_I, c_I in the hazy patch equals the position i_J, j_J, c_J in the clean patch assuming that $\tilde{t}(\mathbf{x})$ is a constant. However, DCP (Eq. (2)) can be applied only when the assumption is satisfied with the conclusion that the darkest pixel in the hazy patch surely corresponds to the darkest pixel in the clean patch. When the assumption could not be satisfied, we can no longer promise that the darkest pixel in the clean patch is selected and Eq. (2) cannot be substituted in Eq. (4). This issue is particularly severe in some edge cases, since the depth of the scene varies from pixel to pixel. Therefore, we calculate the dark channel on both sides of Eq. (3) because $\tilde{t}(\mathbf{x})$ cannot be treated as a constant value.

A simple toy example of vulnerability in DCP. As shown in Figure 3, we present a toy example to demonstrate limitations with the transmission map calculation process in [22]. The four matrices in Figure 3 denote different physical quantities of Eq. (1) in the same local patch, respectively. When calculating the transmission map by [22], the darkest pixel in the hazing image $I(\mathbf{x})$ is used and the final estimation result can be obtained directly by Eq. (5). For the toy example, we calculate the index of the darkest pixel in $I(\mathbf{x})$, which is (3, 2). Substituting this index into Eq. (1), we get

$$I(3, 2) = t(3, 2) * J(3, 2) + 1 - t(3, 2). \quad (8)$$

Using DCP prior (i.e., setting $J(3, 2)$ as zero), we obtain the final estimation result: $t(\mathbf{x}) = 1 - \min(J(\mathbf{x})) = 0.4$, which

overshoots the ground truth $\frac{1}{3}$. The reason for this error is that $J(3, 2)$ is not the darkest pixel in the clean image patch ($J(2, 2) < J(3, 2)$). Consequently, if $J(3, 2) = 0.5$ is regarded zero for the estimation of the transmission map, it violates the assumption $J^{dark} \rightarrow 0$ needed for the DCP model.

To relax such constraints and provide a more precise calculation of the transmission map, we restart from Eq. (3). Following He *et al.* [22], we need to utilize Eq. (2) and make the second term on the right-hand side of Eq. (3) equal to zero. The dark channel prior described in Eq. (2) describes the statistical property of the darkest pixel in a clean image patch. Under the premise that clean-hazy image pairs are available, we can first calculate the index of the darkest pixel in the clean patch and substitute the calculated index into Eq. (3), which can be formulated as:

$$i_{gt}, j_{gt}, c_{gt} = \underset{i, j \in \Omega(\mathbf{x})}{\operatorname{argmin}} \left(\underset{c \in r, g, b}{\operatorname{argmin}} \frac{J^c(i, j)}{A^c} \right). \quad (9)$$

Then, we substitute i_{gt}, j_{gt}, c_{gt} of Eq. (9) into Eq. (3):

$$\frac{I^{c_{gt}}(i_{gt}, j_{gt})}{A^c} = t(i_{gt}, j_{gt}) \frac{J^{c_{gt}}(i_{gt}, j_{gt})}{A^c} + 1 - t(i_{gt}, j_{gt}). \quad (10)$$

In this way, the DCP can be applied based on that the darkest pixel in the clean patch is selected. substituting Eq. (2) into Eq. (10), we can obtain the final estimate of the transmission map as follows:

$$t_{gt}(i_{gt}, j_{gt}) = 1 - \frac{I^{c_{gt}}(i_{gt}, j_{gt})}{A^c}. \quad (11)$$

This formulation calculates the value of the transmission map of the darkest pixel in the clean patch. There is still a need to assume that the transmission map value of the center pixel in the local patch is equal to that of the darkest pixel, which can be formulated as follows:

$$\tilde{t}_{gt}(\mathbf{x}_c) = t_{gt}(i_{gt}, j_{gt}), \quad (12)$$

where $\mathbf{x}_c$ denotes the center pixel of each local patch and (i_{gt}, j_{gt}) denotes the darkest pixel index found in a local patch. As a result, we ease the previous assumption that the transmission map is constant in the entire local patch, and a better calculation of the transmission map can be obtained under the premise that paired data are available. There exists a certain degree of block artifacts in the estimated transmission map by Eq. (12). Here, we adopt the guided filtering [23] proposed by He *et al.* to alleviate this problem.

To example revisited. In contrast, our proposed approach can estimate the transmission map more accurately than the original method developed in [22]. In our estimation process, we first calculate the index of the darkest pixel in the $J(x)$ patch, which is (2, 2). Then we take such an index into Eq. (1) and get:

$$I(2, 2) = t(2, 2) * J(2, 2) + 1 - t(2, 2). \quad (13)$$

Now, we can guarantee that the darkest pixel in $J(x)$ is selected (note that the position of the darkest pixel in $I(x)$ can be inconsistent with that of the darkest pixel in $J(x)$) and the dark channel prior could be used safely. The final estimation result is calculated through Eq. (11), which is $t(x) = 1 - I(2, 2) = 0.3$, closer to the ground truth value ($\frac{1}{3}$) than the result of DCP [22].

B. Reformulation of ASM for Non-homogeneous Haze

In the original atmospheric scattering model, the transmission map has an exact formulation for homogeneous haze:

$$t(\mathbf{x}) = e^{-\beta d(\mathbf{x})}, \quad (14)$$

where β denotes the constant scattering coefficient of the homogeneous haze throughout the scene, and $d(x)$ denotes the depth map. For homogeneous haze, $t(x)$ is only related to one variable (i.e., depth). While in some non-homogeneous scenes, the scattering coefficient varies from pixel to pixel, and $t(x)$ now relates to two variables that are much more complex. We can now reformulate the Eq. (14) as follows:

$$t(\mathbf{x}) = e^{-\beta(\mathbf{x})d(\mathbf{x})}. \quad (15)$$

For non-homogeneous haze, estimating the transmission map correlated to two variables simultaneously could be too hard. To decouple the scattering coefficients from depth values, we first consider the scattering coefficient as a constant and estimate an approximate initial value t_0 using Eq. (11). To take into consideration the change in the scattering coefficient of haze from pixel to pixel, we take the first-order Taylor expansion of the Eq. (15) for the variable $\beta(\mathbf{x})$ and regard $d(x)$ as constant:

$$\begin{aligned} t_\beta(\beta) &= e^{-\beta d} \\ &\approx e^{-\beta_0 d} + \frac{\partial e^{-\beta d}}{\partial \beta} (\beta - \beta_0) + o(\beta - \beta_0) \\ &\approx t_0 + d e^{-\beta_0 d} (\beta - \beta_0) \end{aligned} \quad (16)$$

We denote the second term $d e^{-\beta_0 d} (\beta - \beta_0)$ by $\beta(\mathbf{x})$, and we learn it directly through our network, as shown in Figure 1.

Previous work [59] has also explored the non-homogeneous ASM which considers the spatially-variant scattering coefficient for polarization-based dehazing pipeline. The difference between our reformulation and theirs is that we first observed that estimating the transmission map which simultaneously related to depth and spatially-variant scattering coefficient was too hard. And we try to approximate the final transmission map through first-order Taylor expansion. In [59], they directly decompose the variables in ASM into two parts: the mean value and the spatially-variant turbulence, which is suitable for polarization-based dehazing pipelines but could not fit the normal RGB images well.

C. Uncertainty in the Estimation Transmission Map

With the rise of deep convolution neural networks (DCNN), the idea of estimating ASM-related physical variables (Eq. (1)) via a deep neural network was proposed in the work [19]. In [19], the DCP-estimated transmission map is considered the signal of supervision for training networks, achieving promising results. Following this line of research [19], many follow-up works [31], [9], [36] have been proposed. However, previous work [9] does not address issues such as that the DCP label is not accurate enough, that the haze is often non-homogenous in complex scenes, and that there is uncertainty in the transmission map estimation process.

To address this issue, we propose to model the uncertainty of the data on the transmission map, achieving better performance for the dehazing task than for the baseline method.

1) *Data Uncertainty in transmission map*: In Sec. IV-A, we present how to improve the DCP [22]. Although a more precise transmission map can be calculated, it still leaves a gap between the calculated and the real ones. Therefore, we propose to fill this gap by modeling the uncertainty, called variance, in the transmission map estimation process. In previous works [7], [42], [32], [52], [19] that utilizing the transmission map as a predicted term, MSE or $\mathcal{L}_1$ loss functions are always adopted to measure the distance of transmission map $\hat{t}(x)$ estimated by networks and $\tilde{t}(x)$ calculated by DCP. The probability assumption behind the MSE or $\mathcal{L}_1$ loss is that every pixel obeys the independent and identical distribution with the same variance. Taking $\mathcal{L}_1$ loss as an example, the likelihood of all pixels in a transmission map can be formulated as

$$p(\hat{t}(x)|\tilde{t}(x)) = c \exp\left(-\frac{\|\hat{t}(x) - \tilde{t}(x)\|_1}{\sigma}\right), \quad (17)$$

where c, σ denote spatially invariant constants.

However, such an assumption of stationarity or spatial invariance prior model is invalid for transmission map estimation in the real world, since the ground truth of the transmission map is not available and the dummy labels that are calculated by DCP or enhanced labels that are calculated by the proposed approach are not accurate enough in the wilds. To fill this gap, we propose to model the uncertainty of the transmission map. Specifically, we propose to estimate the transmission map along with the uncertainty, called variance, which can be formulated as follows.

$$p(\hat{t}(x), \sigma(x)|\tilde{t}(x)) = \frac{1}{2\sigma(x)} \exp\left(-\frac{\|\hat{t}(x) - \tilde{t}(x)\|_1}{\sigma(x)}\right), \quad (18)$$

where $\sigma(x)$ denotes the spatial variance of the estimated transmission map by the network, as shown in Figure 1. Then, the log-likelihood can be formulated as follows.

$$\ln p(\hat{t}(x), \sigma(x)|\tilde{t}(x)) = -\frac{\|\hat{t}(x) - \tilde{t}(x)\|_1}{\sigma(x)} - \ln \sigma(x) - \ln 2. \quad (19)$$

For numerical stability, we train the networks to estimate logarithmic variance $\theta(x) = \ln \sigma(x)$. Finally, the maximum likelihood estimate of (19) can be transformed as the minimization of the following loss function to estimate the uncertainty of the transmission map for single image dehazing.

$$\mathcal{L}_{\text{uncert}} = \exp(-\theta(x)) \|\hat{t}(x) - \tilde{t}(x)\|_1 + \theta(x). \quad (20)$$

2) *The Adaptivity of Learned Uncertainty $\hat{t}(x)$* : The learned uncertainty of the transmission map indicates the confidence of the learned transmission map. For example, if the distance between the predicted transmission map and the transmission map label is much larger, $\mathcal{L}_{\text{uncert}}$ (20) will intend to have a larger variance $\theta(x)$, which is also called uncertainty in this paper to lower the value of the first terms. On the one hand, $\mathcal{L}_{\text{uncert}}$ will not generate a large variance $\theta(x)$ for the entire sample, for the second term of Eq. (20) penalizes the

model in turn. On the other hand, $\mathcal{L}_{\text{uncert}}$ will not generate a small variance $\theta(x)$ for all samples, causing the first term of Eq. (20) will punish the model in return. Therefore, it is not necessary to prepare the uncertainty label $\hat{t}(x)$. The uncertainty $\hat{t}(x)$ can be adaptively learned by Eq. (20) with suitable values.

D. Physics Model-Based Network

The general framework of our network, as shown in Figure 1, is composed of two parts: the neural network of the backbone and four different heads for estimating physical variables in ASM (Eq. (1)). When a hazy image is fed into our network, its corresponding physical variables (i.e., $\mathbf{A}$, $\mathbf{J}(x)$ and $t(x)$) in the ASM of Eq. (1), $\theta(x)$ and $\beta(x)$ (non-homogeneous only) will be estimated.

Following [19], we select FFANet [40] as our backbone network in our proposed framework. The structure of the backbone network is shown in Figure 2, and the details of the network in our implementation will be described in detail in Section IV-A. The input of the backbone network is the original hazy images. After the complex computation of convolution and the attention mechanism, feature maps that contain rich spatial information can be extracted. After that, four individual heads with the same architecture are adopted to estimate physical variables such as clean image $\mathbf{J}(x)$, transmission map $t(x)$, uncertainty of transmission map $\theta(x)$ and $\beta(x)$ (non-homogeneous only). To calculate atmospheric light $\mathbf{A}$, DCPDN [52] is used and acts as an independent part of the backbone network.

$$\hat{\mathbf{J}}(x), \hat{t}(x), \theta(x), \hat{\mathbf{A}}, \beta(x) = \mathcal{F}_{\text{cnn}}(\mathbf{I}(x)), \quad (21)$$

where $\hat{\mathbf{J}}(x)$ denotes the output of clean image, $\hat{t}(x)$ denotes the estimated transmission map by network, $\theta(x) = \ln \sigma(x)$ denotes the estimated uncertainty of the transmission map, $\hat{\mathbf{A}}$ denotes the estimated global atmosphere light, $\beta(x)$ denotes the term in Eq. (16), and $\mathcal{F}_{\text{cnn}}$ denote deep-convolution neural networks.

E. Overall Loss Function

The overall loss function of our method consists of three parts. The first one is L_1 loss to constrain the output of a clean image to be close to the hazy-free image label, which can be expressed as:

$$\mathcal{L}_{\text{clean}} = \|\mathbf{J}(x) - \hat{\mathbf{J}}(x)\|_1. \quad (22)$$

To fully utilize the physical model ASM, we reconstruct the hazy image again with estimated physical variables such as transmission map $\hat{t}(x)$ and atmospheric light $\hat{\mathbf{A}}$ and an estimated clean image $\hat{\mathbf{J}}(x)$ based on ASM (Eq. (1)). The reconstruction process can be expressed as: $\hat{\mathbf{I}}(x) = \hat{\mathbf{J}}(x) \odot \hat{t}(x) + \hat{\mathbf{A}} \odot (1 - \hat{t}(x))$. Then, the L_1 loss is adopted to measure the distance between the reconstructed hazy image and the input one. The loss function can be formulated as follows.

$$\mathcal{L}_{\text{haze}} = \|\mathbf{I}(x) - \hat{\mathbf{I}}(x)\|_1. \quad (23)$$

Finally, we introduce the uncertainty loss (Eq. (20)) that has been derived in Section III-C into the overall loss function. The final loss function can be formulated as follows:

$$\mathcal{L}_{all} = \lambda_{clean}\mathcal{L}_{clean} + \lambda_{haze}\mathcal{L}_{haze} + \lambda_{uncert}\mathcal{L}_{uncert}, \quad (24)$$

where λ_{clean} , λ_{haze} and λ_{uncert} are the trade-off hyperparameters.

IV. EXPERIMENTAL RESULTS

A. Experimental Setups

Datasets and Metrics. Both synthetic and real-world datasets are used to evaluate our proposed method. The widely used synthetic image dehazing dataset RESIDE- β [33] consists of five subsets: Indoor Training Set (ITS), Outdoor Training Set (OTS), Synthetic Objective Testing Set (SOTS), Real-world Task-Driven Testing Set (RTTS) and Hybrid Subjective Testing Set (HSTS). Following [49], we select ITS as our training dataset and evaluate our proposed method on the synthetic SOTS-indoor dataset. To further demonstrate the effectiveness in real-world scenes, we also adopted two real-world datasets. Dense-Haze [1] and NH-HAZE [2]. To compare our method with previous state-of-the-art methods, the widely used metrics of single image dehazing, Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) are adopted to evaluate the performance of our approach.

Implementation Details. To augment the training dataset, we randomly rotate the image by 90° , 180° , 270° , and horizontally flip. The hazy image is cropped with the size 256×256 as input of our model in SOTS-indoor and 512×512 in Dense-Haze and NH-HAZE. The whole network is trained by 3×10^5 steps on the synthetic dataset and 1×10^5 for the real datasets. The ADAM algorithm [29] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ is adopted to optimize the network. The initial learning rate is set to 1×10^{-4} , and the cosine annealing strategy is adopted to adjust the learning rate from the initial learning rate to zero. Regarding the hyperparameters λ_{clean} , λ_{haze} and λ_{uncert} , they are set as 0.1, 0.01, 0.001 respectively to achieve a good balance of three terms. Our method is implemented by PyTorch 1.7.0 and trained on an NVIDIA RTX 3090Ti GPU.

For the SOTS-indoor dataset containing the ground truth transmission map that corresponds to hazy images, we directly supervise the output $\hat{t}(x)$ of our model with the ground truth transmission map provided. Note that SOTS-indoor is a homogeneous dehazing dataset, so the final transmission map is $\hat{t}(x)$ as shown in Figure 1. For Dense-Haze and NH-HAZE, it is almost impossible to obtain the accurate ground truth of the transmission map. Thus, we adopt the improved transmission map that is calculated by the proposed approach through Eq. (12) as the supervised signal to learn the transmission map by networks. Also, the two real-world datasets are non-homogeneous dehazing datasets. According to Eq. (16), the final transmission map is the sum of $\hat{t}(x)$ and $\beta(x)$ as discussed in Sec. III-B.

Network Details. Following the network configuration of the original FFANet [40]. As shown in Figure 2, the number of the group structure G is set as 3, and the number of the

basic block structure B is 19. After obtaining the feature map calculated by the backbone network, we adopt four individual heads to produce the final outputs in Eq. (1), where the Beta head is only used for non-homogeneous dehazing datasets. Each of the heads consists of three convolution layers that are without bias parameters, the channels of which are 64, 32, and 3, respectively. Between the first convolution and the second convolution, we add a ReLU activation function.

B. Overall Performance

Results on Synthetic homogeneous Dataset. In Table I, we compare our performance with some state-of-the-art methods on the SOTS-indoor [33] dataset, which is a synthetic homogeneous dehazing dataset. For the indoor synthetic dehazing dataset, it is simple to acquire the overall depth map of the entire scene. As a result, the transmission map provided by the dataset is precise enough and could be used directly. Under such conditions, we directly adopt the output $T(x)$ of our network as the final estimate of the transmission map. Our method achieves the best performance with 39.26dB PSNR and 0.9916 SSIM, outperforming more than 2dB PSNR as the second top performance. The dehazed performance on the SOTS-indoor dataset shows that our method is more effective and gains 0.85dB more than [50] and 0.64dB more than [25]. Note that the backbone of our proposed method is FFA-Net [40]. Our proposed method achieves a large gain, which is **2.87dB** comparing FFA-Net. Visual comparisons of the restored image quality on the ITS dataset are presented in Figure 4, Figure 5, and Figure 6. Compared to DCP, FFA-Net [40], MSBDN [13], AECR-Net [49], and Dehamer [20] achieve better dehazed performance. However, there still remains little haze artifacts, as shown in the two green anchors in Figure 1. Our method produces the most visually comfortable results among all methods.

Results on Real-world non-homogeneous Datasets. We also perform our method on real-world datasets: Dense-Haze [1] and NH-HAZE [2], which are two non-homogeneous hazy datasets and compare the final performance with some SOTA methods. It is much more difficult to achieve the same dehazing performance on real-world datasets than the synthetic dataset. This is due to the complex degradation process of the real haze, whereas the synthetic dehazing dataset treats the haze as homogeneous. In addition, it is almost impossible to obtain an exact accurate depth map of the outdoor scenes, which is especially troublesome for physics-based dehazing algorithms. To better handle the complex scene, we follow the method proposed in Sec. III-B and add the output $T(x)$ and $\beta(x)$ together as the final estimation of the transmission map.

For Dense-haze [1], the whole scene is in a severely bad atmosphere, and could not be distinguished through our eyes. But our method successfully improves the visual quality. As shown in Table 1, our method achieves the new SOTA performance on the difficult Dense-Haze dataset. Comparing backbone method FFA-Net, our proposed non-homogeneous version can gain **4.69dB** and **0.9dB** improvement in Dense-Haze and NH-HAZE datasets respectively. Comparing state-

TABLE I: The average PSNR and SSIM dehazing results comparisons on three different benchmark datasets. The best performance is shown in **bold** and the second-best performance is shown in underline.

Methods	SOTS-indoor [33]		Dense-Haze [1]		NH-HAZE [2]		Params
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	
(TPAMI'10) DCP [22]	15.09	0.7649	10.06	0.3856	10.57	0.5196	-
(ICCV'19)GridDehazeNet [35]	32.16	0.9836	13.31	0.3681	13.80	0.5370	0.96M
(AAAI'20)FFA-Net [40]	36.39	0.9886	12.22	0.4440	18.13	0.6473	4.68M
(CVPR'20)MSBDN [13]	33.79	0.9840	15.37	0.4858	19.23	<u>0.7056</u>	31.35M
(CVPR'21)UHD [58]	21.75	0.8786	12.16	0.4594	16.05	0.4612	34.54M
(CVPR'21)AECR-Net [49]	37.17	0.9901	15.80	0.4660	<u>19.88</u>	0.7173	2.61M
(CVPR'22)DeHamer [20]	36.63	0.9881	16.62	0.5602	20.66	0.6844	132.44M
(ECCV'22) PMNet [50]	38.41	0.9900	-	-	-	-	-
(ECCV'22) BILD [25]	38.58	0.9921	-	-	-	-	-
(AAAI'22) UDN [24]	<u>38.62</u>	0.9909	-	-	-	-	4.25M
Ours (homogeneous)	39.26	<u>0.9916</u>	16.66	0.5730	18.60	0.6440	6.32M
Ours (non-homogeneous)	-	-	16.91	<u>0.5710</u>	19.03	0.6503	6.38M



Fig. 4: Single image dehazing visual results comparison on dataset SOTS-indoor [33]. (a) input, (b) DCP [22], (c) GridDehazeNet [35], (d) FFA-Net [40], (e) MSBDN [13], (f) AECR-Net [49], (g) UHD [58], (h) DeHamer [20], (i) ours, (j) Ground-truth.

of-the-art method DeHamer [20], our proposed method is inferior to DeHamer [20] on the NH-HAZE dataset. But note that the parameters of DeHamer [20] is **132.44M**, which is over **20 times** of proposed method. For NH-HAZE, our method is inferior to other competing methods, probably because the backbone used in our method is relatively simple and cannot handle complex real haze. When compared with the backbone network FFA-Net [40], the proposed method achieved 0.9 dB gain over FFA-Net [40] for NH-HAZE dataset. The visual results of the test methods for Dense-haze and NH-HAZE datasets are shown in Figure 7, and 8, from which one can see that our method produced visually pleasant results.

C. Ablation Study

Effectiveness of the improved DCP.

To further demonstrate the effectiveness of our improved DCP in Sec. IV-A, we directly compare the final dehazing performance with DCP [22] on the Dense-Haze dataset. We first utilize the transmission map estimation method proposed in [22] and adopt guided filtering to refine the final result. This intermediate result is denoted as $T_0(x)$ and serves as the supervision signal of the output $T(x)$. Then we estimate

TABLE II: Comparison of DCP and our refined DCP on the Dense-Haze dataset.

Methods	Dense-Haze	
	PSNR	SSIM
DCP [22]	15.58	0.5594
Ours	16.91	0.5710

the transmission map in our proposed method through Eq. (12) along with the guided filtering to refine the estimation result. Improved supervision of the transmission map can lead to better PSNR performance, which can be seen from Table II. The final result shows that our proposed method to estimate the transmission map can achieve better performance.

Ablation study of different weights in the loss function.

To explore the effectiveness of the different elements in our loss function and the best weight for each element, we performed an ablation study to analyze the contribution of each. We try different weights in Eq. (24), and the results are presented in Table III: we first adopt only $\mathcal{L}_{clean}$ which aims to reconstruct the clean image and the whole work is just the same as FFA-Net [40]. The coefficient of $\mathcal{L}_{clean}$ is set to 1 and the final performance is 36.39dB. Then we

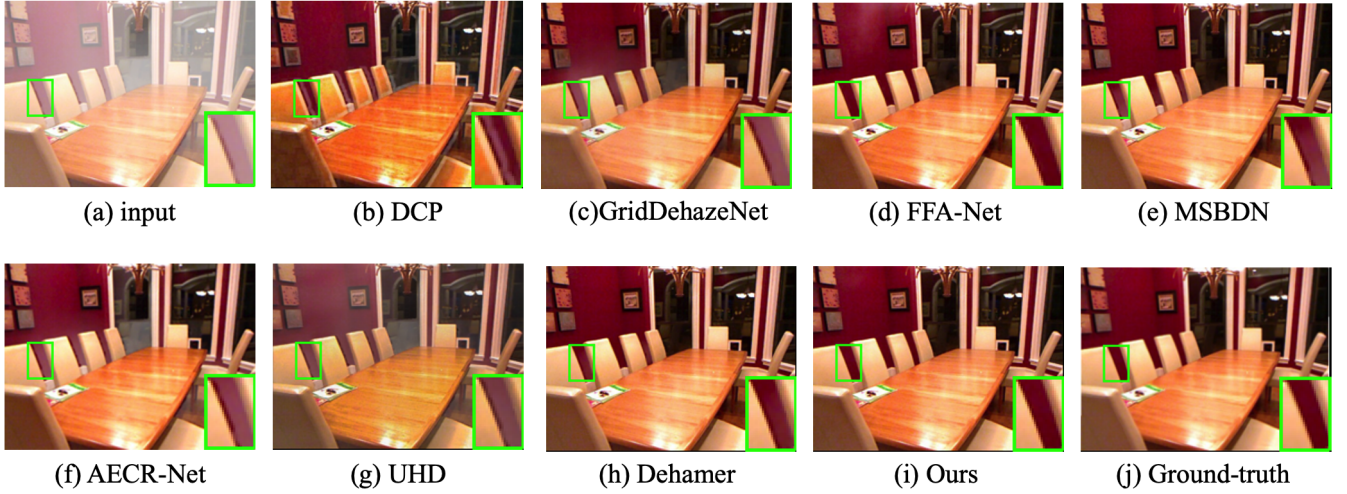


Fig. 5: Single image dehazing visual results comparison on dataset SOTS-indoor [33]. (a) input, (b) DCP [22], (c) GridDehazeNet [35], (d) FFA-Net [40], (e) MSBDN [13], (f) AECR-Net [49], (g) UHD [58], (h) Dehamer [20], (i) ours, (j) Ground-truth.

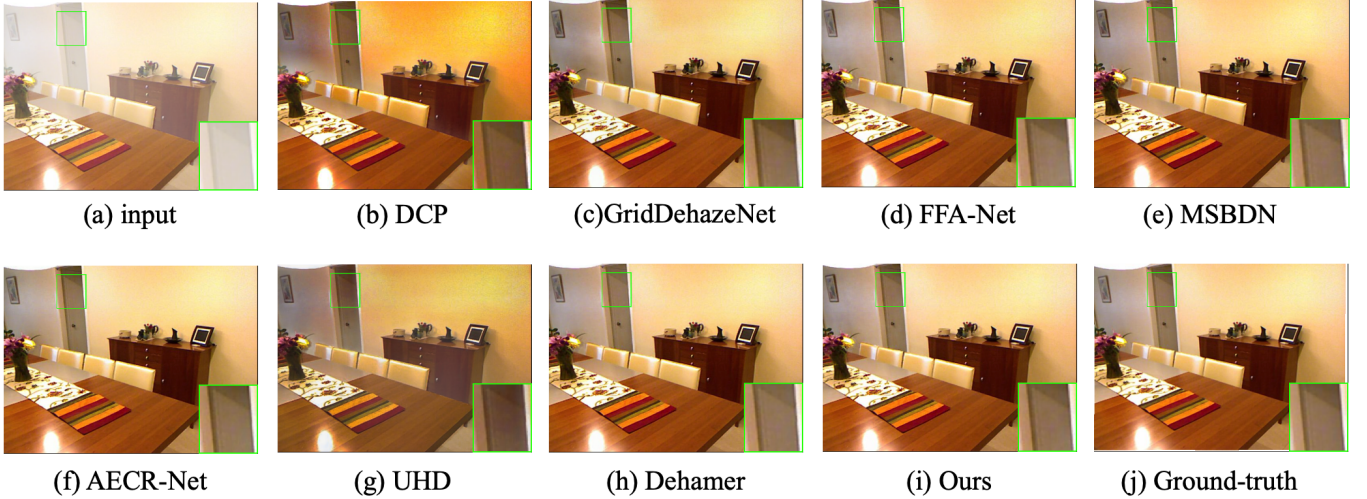


Fig. 6: Single image dehazing visual results comparison on dataset SOTS-indoor [33]. (a) input, (b) DCP [22], (c) GridDehazeNet [35], (d) FFA-Net [40], (e) MSBDN [13], (f) AECR-Net [49], (g) UHD [58], (h) Dehamer [20], (i) ours, (j) Ground-truth.

TABLE III: Ablation study of each part in the overall loss function on the SOTS-indoor dataset.

λ_{clean}	λ_{haze}	λ_{uncert}	PSNR	SSIM
1	0	0	36.39	0.9886
1	1	0	31.06	0.9689
1	0.2	0	35.23	0.9867
1	0.1	0	36.47	0.9879
1	0.1	0.03	35.16	0.9863
1	0.1	0.008	36.35	0.9883
1	0.1	0.01	39.26	0.9916

adjust the weights of the cycle loss $\mathcal{L}_{haze}$, which aims to help the network understand the exact physical meaning of each output in ASM. The reconstruction process of the hazy image forces the network to produce a result that could fit in the physical model ASM. The best performance is 36.47dB which

corresponds to a weight of 0.1. And this shows that $\mathcal{L}_{haze}$ improves the quality of the dehazing. Finally, we model the transmission map estimation process with uncertainty $\mathcal{L}_{uncert}$, which corresponds to the final row in Table III. The best coefficient of uncertainty loss is 0.01. The $\mathcal{L}_{uncert}$ greatly improves the dehazing performance and achieves 39.26dB. The result in another way proves the importance of estimation transmission in the model-based dehazing process. Finally, the best combination of the three weights is 1, 0.1, and 0.01.

Effectiveness of the Reformulation of the ASM for the non-homogeneous dataset. We conduct an ablation study to examine whether our analysis of the transmission map for the non-homogeneous hazy image is correct. We first treat non-homogeneous hazy images the same as homogeneous hazy images, and directly adopt the output $\hat{t}(x)$ of our network as

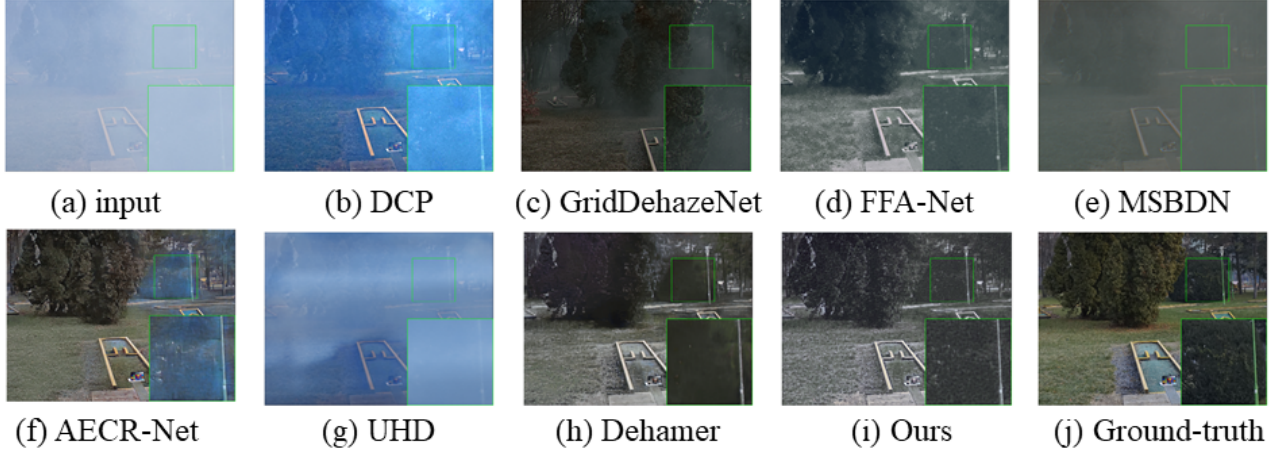


Fig. 7: Single image dehazing visual results comparison on Dense-Haze dataset [1]. (a) input, (b) DCP [22], (c) GridDehazeNet [35], (d) FFA-Net [40], (e) MSBDN [13], (f) AECR-Net [49], (g) UHD [58], (h) Dehamer [20], (i) ours, (j) Ground-truth.

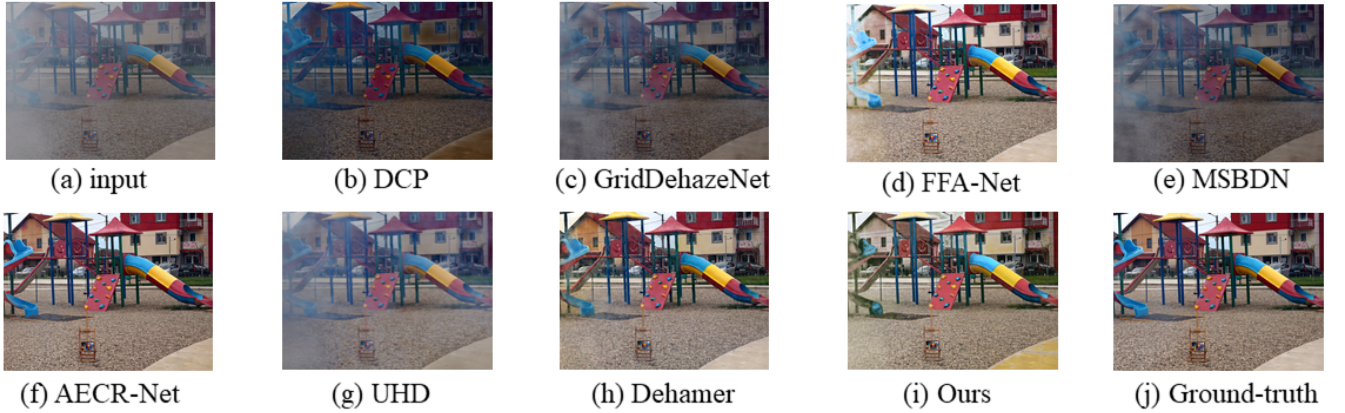


Fig. 8: Single image dehazing visual results comparison on dataset NH-Haze [2]. (a) input, (b) DCP [22], (c) GridDehazeNet [35], (d) FFA-Net [40], (e) MSBDN [13], (f) AECR-Net [49], (g) UHD [58], (h) Dehamer [20], (i) ours, (j) Ground-truth.

TABLE IV: Ablation study of the reformulation of ASM for the non-homogeneous hazy dataset.

Methods	Dense-Haze		NH-HAZE		Params
	PSNR	SSIM	PSNR	SSIM	
Original Formulation	16.66	0.5730	18.60	0.6440	6.32M
Reformulation	16.91	0.5710	19.03	0.6503	6.38M

the final estimation of the transmission map. Next, we follow the method proposed in Sec. III-B, and add the output $\hat{t}(x)$ and $\beta(x)$ together as the final estimation of the transmission map. The result of the two experiments is shown in Table IV. Comparing our proposed homogeneous modeling, our proposed non-homogeneous modeling can bring near **0.3dB** on non-homogeneous dataset of Dense-Haze and NH-HAZE dataset, which also demonstrates the effectiveness of proposed non-homogeneous modeling. Our non-homogeneous method can further improve the performance in non-homogeneous dataset.

D. Visualization of Estimated $t(x)$ and $\beta(x)$

We have visualized the estimated transmission map and $\beta(x)$ in Fig. 9. For homogeneous haze, the estimated transmission map reflects the depth map of the scene as Eq. (15) described. The deeper pixels intend to have smaller values of the transmission map. The learned feature $\beta(x)$ reflects the uncertainty of the estimated transmission map for each pixel, which has a larger value for the long-distance pixels and a smaller value for the closer pixels. The estimated transmission map of closer pixels is more accurate than the long-distance pixels since the transmission map of long-distance pixels is harder to estimate accurately. Therefore, the uncertainty of the estimated transmission map is bigger than the closer ones.

E. Implications

Automatic driving on hazy days could be quite a problem. The video captured by the camera is distorted by floating particles, which makes the object detection algorithm not work. Moreover, the haze could be non-homogeneous, which

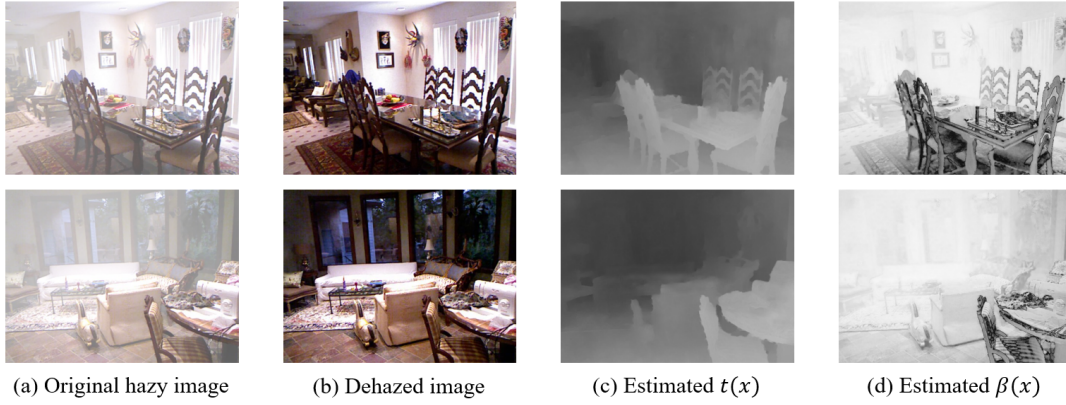


Fig. 9: Visualization of the estimated $t(x)$ and learnt feature $\beta(x)$

is more complex for dehazing methods. Our method could handle complex hazy scenes and perform well for dehazing. The dehazed video is more visually pleasing and could be handled by the auto-driving algorithm. Our method could be utilized in rescuing at sea and understanding the urban road scene on hazy days.

V. CONCLUSION

In this paper, we analyze the problem in the estimation process in DCP and propose to improve the estimation result under the premise that hazy-clean image pairs are available. We combine deep learning and DCP in such a way that the intermediate output transmission map can serve as a self-supervision signal to improve the existing DCP. To address the challenge with non-homogeneous haze, we have extended the conventional physical model by making scattering coefficients spatially varying. Moreover, non-homogeneous haze inspired us to introduce uncertainty modeling into the estimation of the transmission map, which is well aligned with the emerging Bayesian deep learning framework. We have uncertainty improves the flexibility of the network. We have comprehensively evaluated our proposed method on both synthetic and real-world datasets. The result proves the effectiveness of our method.

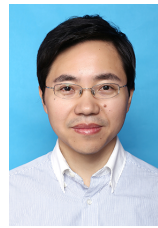
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